MALNAD COLLEGE OF ENGINEERING

[An Autonomous Institution under VTU, Belagavi]

Hassan, Karnataka 573202, India

Department of Computer Science and Engineering

Course Name: **OBJECT ORIENTED PROGRAMMING WITH JAVA**

Course Code: **22CS307A**

Semester: **3rd Semester**

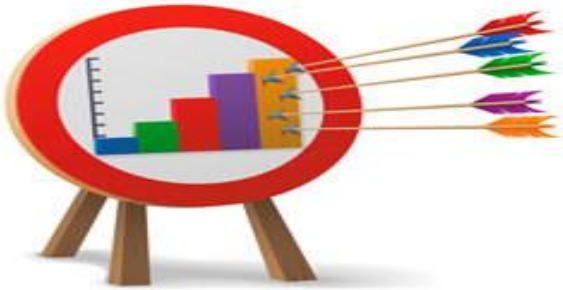
Course Faculty: **Sumanth C M,**

Asst. Professor,

Dept. of CS&E, MCE

Course Objective

Design and develop java application programs using object-oriented concepts.

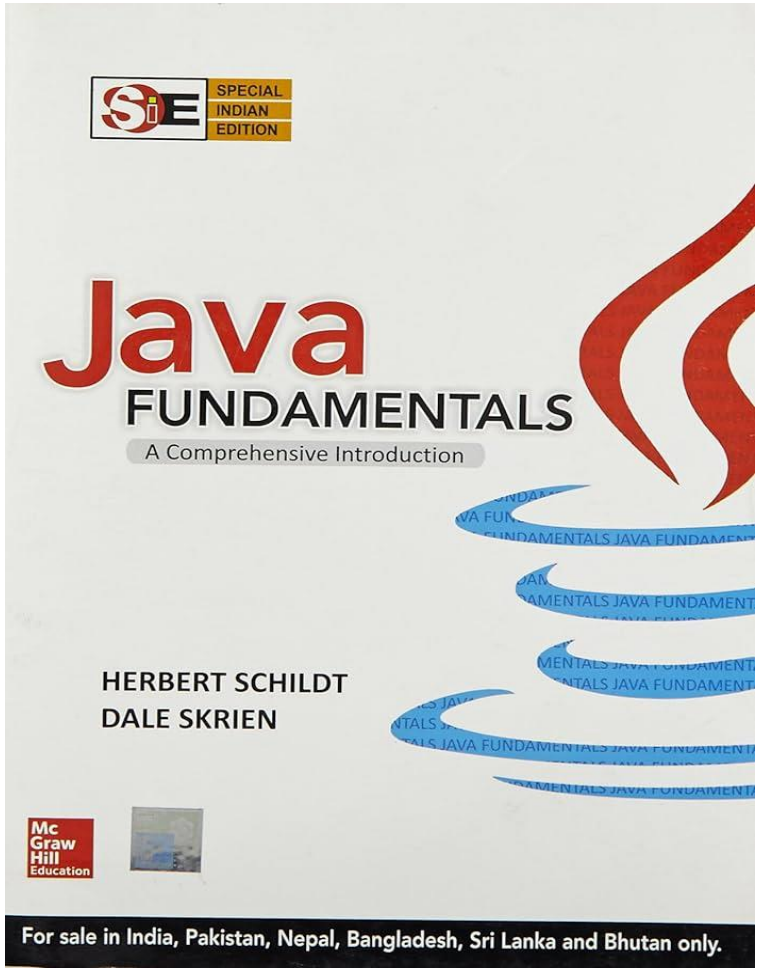


Course Outcome

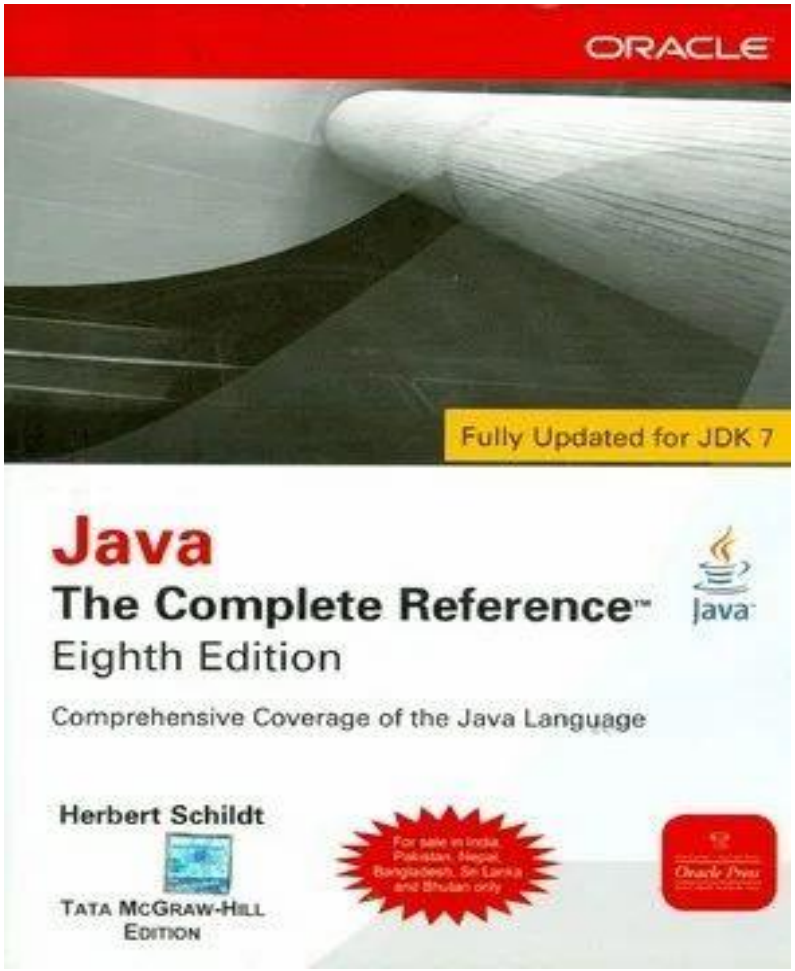
CO	Statement
1	Apply Java constructs for the development of object oriented programs.
2	Analyse the given java program to make suitable changes.
3	Design a java program for the given problem.
4	Conduct practical experiments for demonstrating object oriented concepts through java using IDE.

OBJECT ORIENTED PROGRAMMING WITH JAVA (22CS307A)

Textbooks

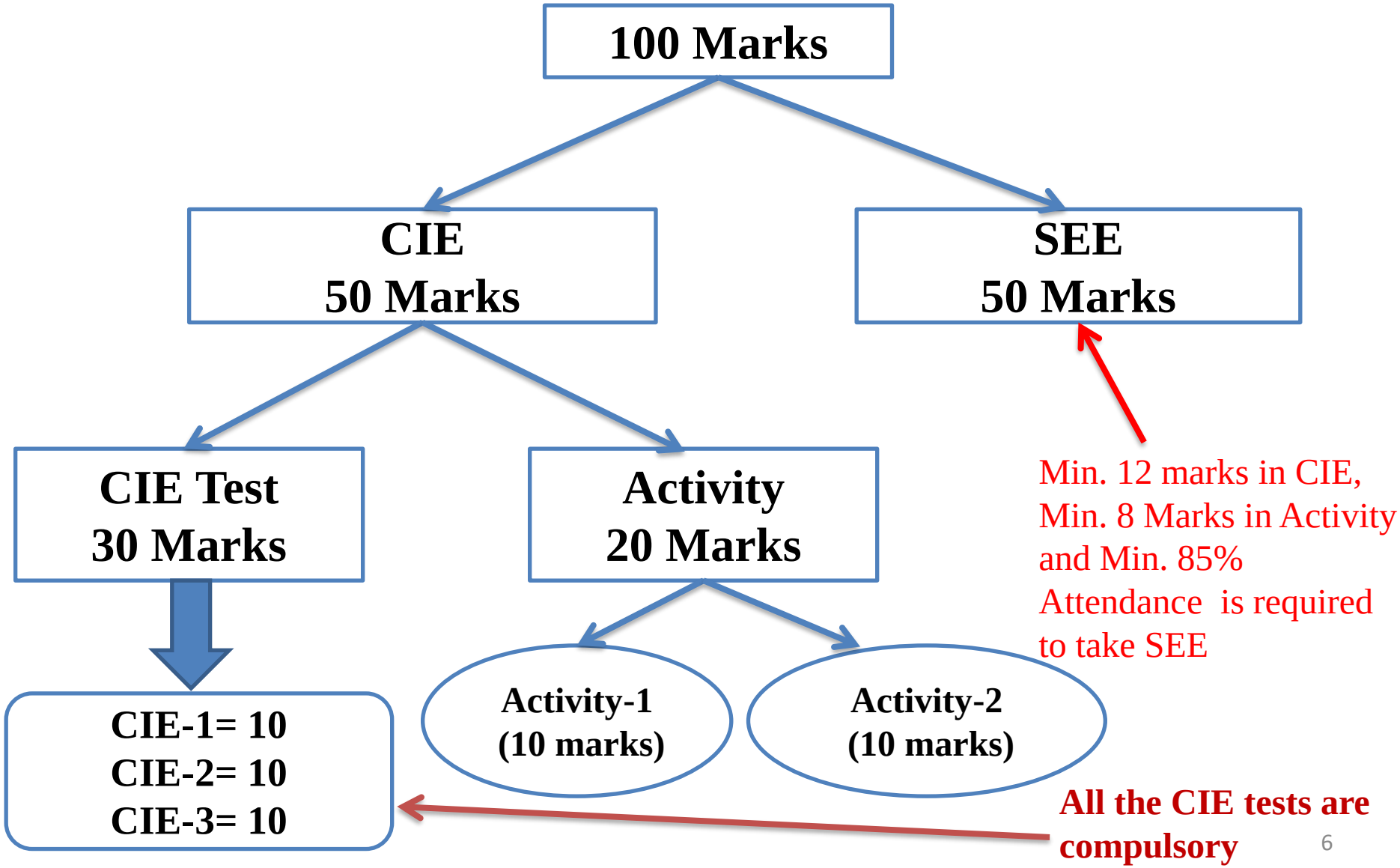


Java Fundamentals, A comprehensive Introduction by Herbert Schildt, Dale Skrien. Tata McGrawHill Edition 2013 (Chapters 1,2,3,4,5,6,7,8,9,10,12)



Java The complete Reference, by Herbert Schildt, Eight Edition Tata Mcgraw Hill Education (Chapter 19).

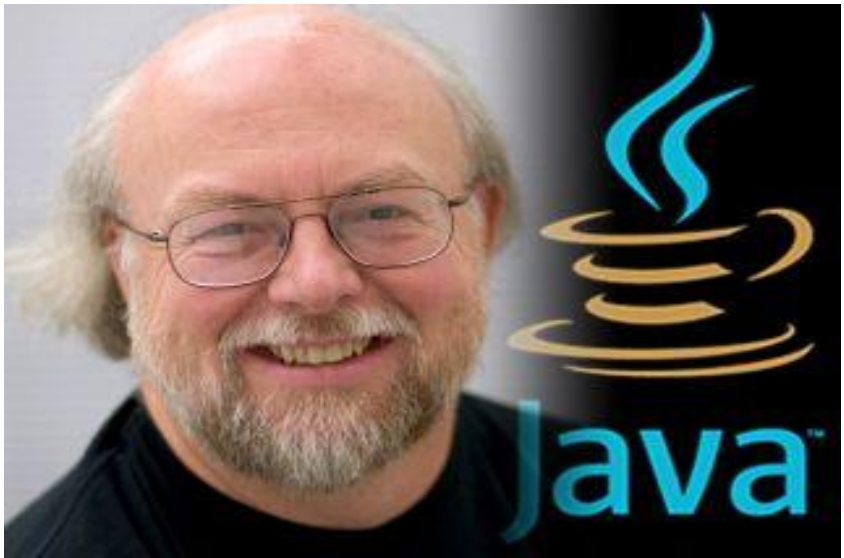
Assessment


100 Marks
CIE
50 Marks
SEE
50 Marks
CIE Test
30 Marks
Activity
20 Marks
Min. 12 marks in CIE, Min. 8 Marks in Activity and Min. 85% Attendance is required to take SEE
CIE-1= 10
CIE-2= 10
CIE-3= 10
Activity-1
(10 marks)
Activity-2
(10 marks)
All the CIE tests are compulsory

What is Java?

- Java is a **high-level**, **general-purpose**, **object-oriented**, and **secure** programming language and platform.
- Java language was developed by **James Gosling** at **Sun Microsystems, Inc. in 1991.**
- It is formally known as **OAK**.
- In 1995, Sun Microsystem changed the name to Java.
- In 2009, Sun Microsystem takeover by Oracle Corporation.

- Java is a platform-independent language- any hardware or software that allows us to run different application on it we call it as platform.



Editions of Java

- There are mainly 3 editions of Java that are available today:

Cont..

- **Java Standard Editions (SE):** It is used to create programs for a desktop computer.
- **Java Enterprise Edition (EE):** It is used to create large programs that run on the server and manages heavy traffic and complex transactions.
- **Java Micro Edition (ME):** It is used to develop applications for small devices such as set-top boxes, phone, and appliances.

Java Timeline

Java SE Version	Version Number	Release Date
JDK 1.0 (Oak)	1.0	January 1996
JDK 1.1	1.1	February 1997
J2SE 1.2 (Playground)	1.2	December 1998
J2SE 1.3 (Kestrel)	1.3	May 2000
J2SE 1.4 (Merlin)	1.4	February 2002
J2SE 5.0 (Tiger)	1.5	September 2004
Java SE 6 (Mustang)	1.6	December 2006
Java SE 7 (Dolphin)	1.7	July 2011
Java SE 8	1.8	March 2014
Java SE 9	9	September, 21st 2017
Java SE 10	10	March, 20th 2018
Java SE 11	11	September, 25th 2018
Java SE 12	12	March, 19th 2019
Java SE 13	13	September, 17th 2019
Java SE 14	14	March, 17th 2020
Java SE 15	15	September, 15th 2020
Java SE 16	16	March, 16th 2021
Java SE 17	17	September, 14th 2021
Java SE 18	18	March, 22nd 2022
Java SE 19		September, 20th 2022
Java SE 20	20	March, 21st 2023
Java SE 21	21	September, 19th 2023

Features of Java

Simple

Portable

Object-oriented

Secure

Distributed

Dynamic

Robust

High Performance

Features of Java

Feature of Java

- Simple
- Portable
- Object-oriented
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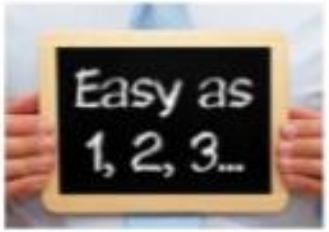
- High Performance

➢ Java was designed to be easy for professional programmer to learn and use effectively.



Simple

Features of Java

Feature of Java

- Simple
- **Portable**
- Object-oriented
- Secure
- Distributed
- Dynamic
- Robust
- High Performance

➢ Applications written using Java are portable in the sense that they can be executed on any kind of computer containing any CPU or any operating system.

Features of Java

Feature of Java

- Simple
- Portable
- **Object-oriented**
- Secure
- Distributed
- Dynamic
- Robust
- High Performance

- Java is true object oriented language.
- Everything is considered to be an "object" and all operations are performed using these objects.

Features of Java

Feature of Java

- Simple
- Portable
- Object-oriented
- **Secure**
- Distributed
- Dynamic
- Robust
- High Performance

- Java Programs run inside virtual machine sandbox to prevent any activity from untrusted sources.
- No use of explicit pointer.

Features of Java

Feature of Java

- Simple
- Portable
- Object-oriented
- Secure
- **Distributed**
- Dynamic
- Robust
- High Performance

- Java has a feature called Remote Method Invocation (RMI) using which a program can invoke method of another program across a network and get the output.

Distributed

Features of Java

Feature of Java

- Simple
- Portable
- Object-oriented
- Secure
- Distributed
- **Dynamic**
- Robust
- High Performance

- Java programs carry with them substantial amounts of run-time type information that is used to verify and resolve accesses to objects at run time.

Features of Java

Feature of Java

- Simple
- Portable
- Object-oriented
- Secure
- Distributed
- Dynamic
- **Robust**
- High Performance

- Java checks the code during the compilation time and run time also.
- Java completely takes care of memory allocation and releasing, which makes the Java program more robust.

Features of Java

Feature of Java

- Simple
- Portable
- Object-oriented
- Secure
- Distributed
- Dynamic
- Robust
- **High Performance**

➢ Java achieves high performance through the use of bytecode which can be easily translated into native machine code.

Java Platform

- A platform is a **software or hardware environment in which a program executes or runs.** Some of the most platforms are Microsoft Windows, Linux, Mac, and Solaris.
- Java platform is only a software-based platform that runs on top of other hardware-based platforms such as Microsoft Windows, Linux, and Mac.

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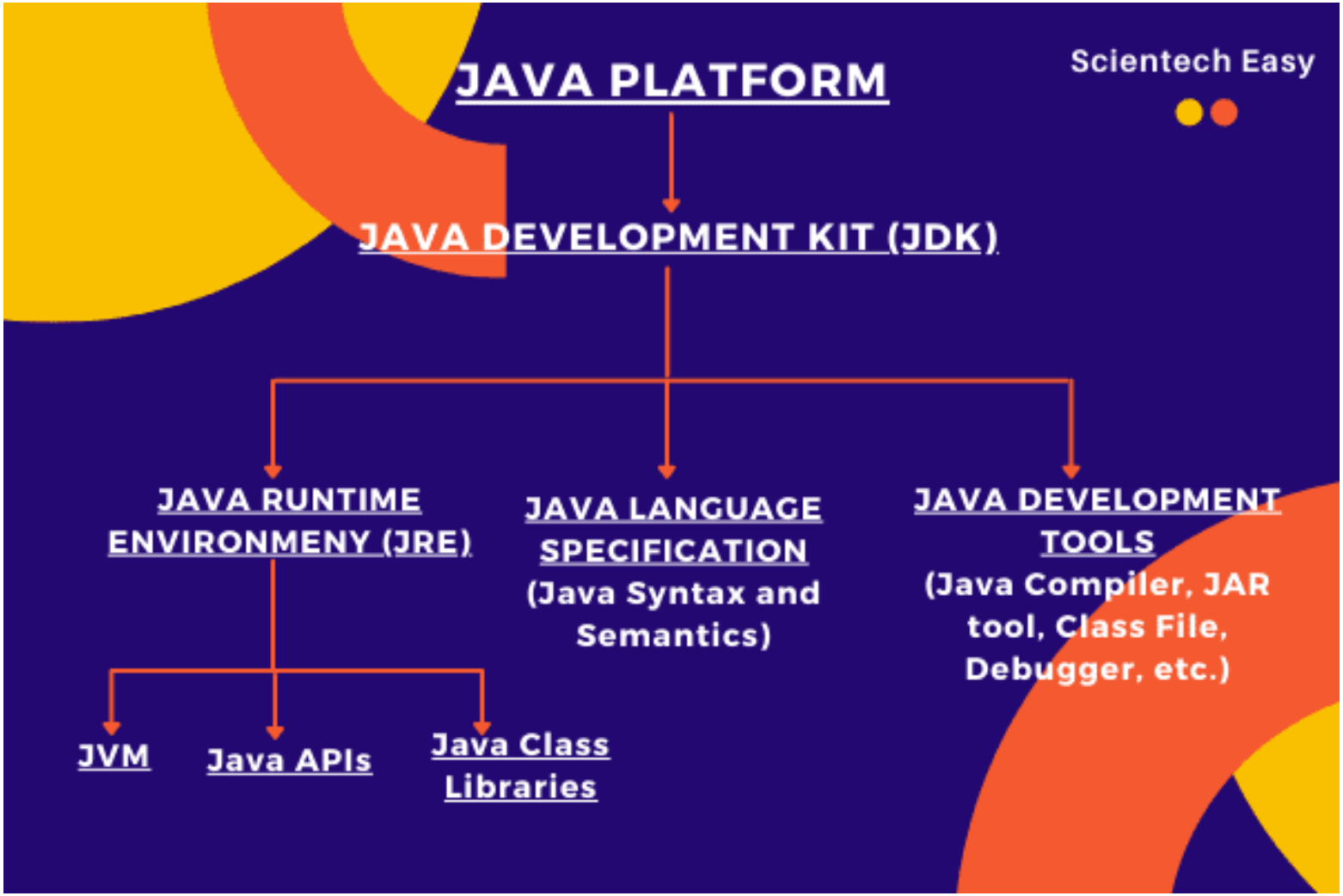
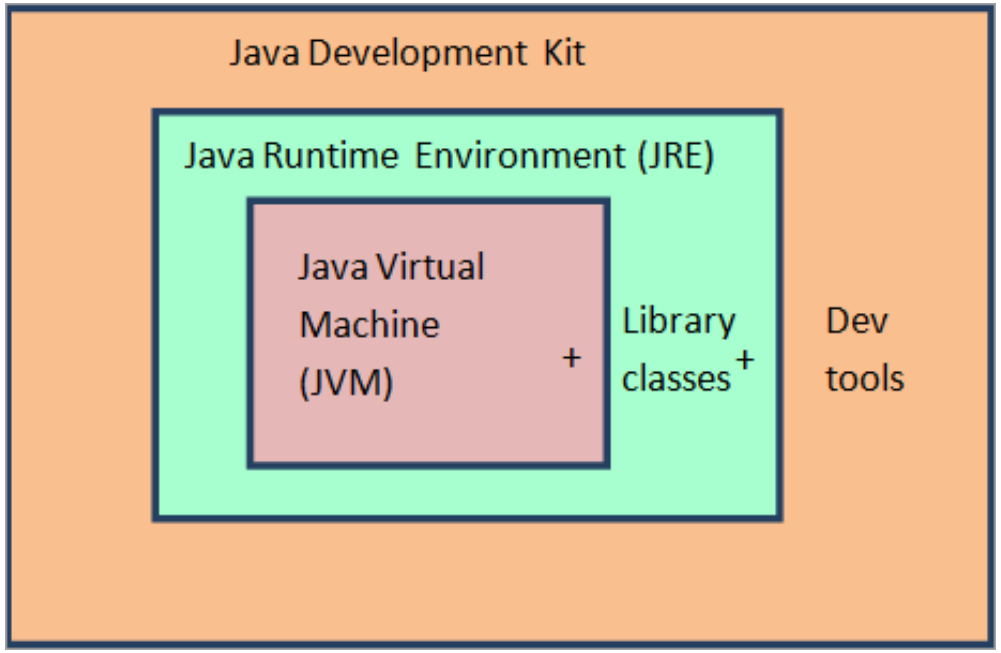
- Java Platform is composed of a hierarchy of components that is shown in the below figure.



What is Java Development Kit (JDK)?

- **Java Development Kit** (JDK) is a software development environment that is developed and distributed by Oracle. It is used for building (developing) java software applications.
- In simple words, it is used to write java programs by programmers.
- Java Development Kit has the following components. They are as follows:



Cont..

Java Runtime Environment (JRE):

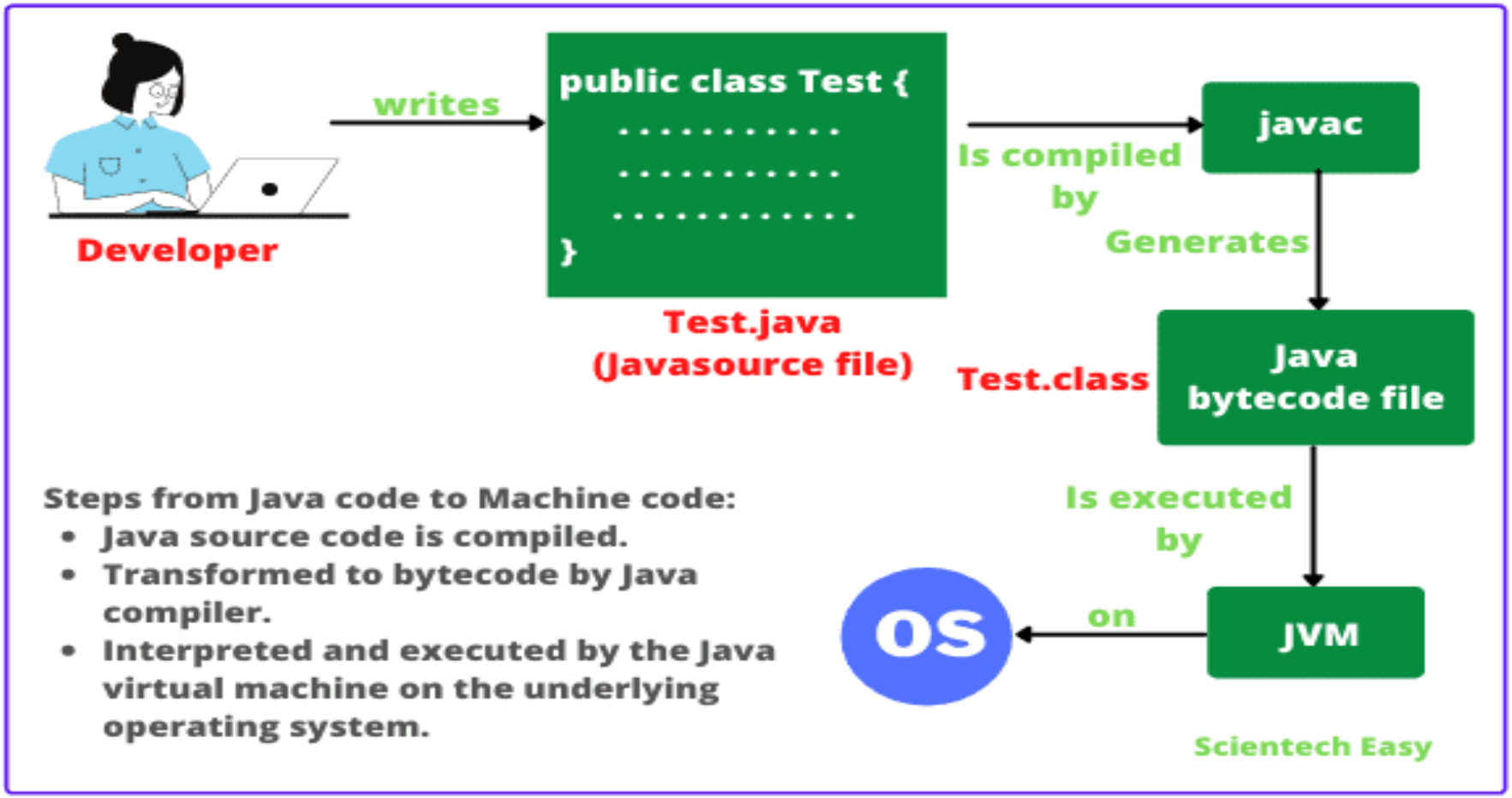
- It is an environment that is required for running (executing) a java application.
- It cannot be used for the development of Java applications.
- It is used only for executing the application program.
- Java Runtime Environment is a collection of programs that contain JVM (Java Virtual Machine), Java APIs (Application Programming Interface), and Java Class Libraries that help in the development of Java application programs.

Cont..

- **Java Virtual Machine (JVM) in Java** is the heart of the entire execution process of the Java program.
- It is basically a program that **provides the runtime environment necessary for Java programs to execute.**
- The specification of JVM is provided by *Sun Microsystem* whose implementation provides a runtime environment to execute our Java applications. JVM implementation is known as Java Runtime Environment (JRE).
- Since **JVM is platform-dependent**, therefore, it is available for many hardware and software platforms.
- The most popular JVM is **HotSpot** that is produced by Oracle. It is available for many operating systems such as Windows, Linux, Solaris, and Mac OS.
- We cannot run the Java program unless JVM is available for the appropriate hardware and operating system platform.

Java Virtual Machine performs the following operations for execution of the program. They are as follows:

a) Load the code into memory.
b) Verifies the code.
c) Executes the code
d) Provides runtime environment.



Bytecode in Java

- **Bytecode in Java** is a highly optimized set of instructions for the *Java Virtual Machine* (JVM) that reads and interprets to run the java program.
- A bytecode is a **binary program code that can only run on JVM.**
- In other words, it is a **machine language (code) for JVM in the form of .class** file, but it is not machine specific because it is not a native code.
- Java bytecode has two most important characteristics that are as follows:
  - **Byte code is independent of processor**, i.e., Java program can be executed on any processor architecture.
  - **It does not depend on operating systems such as Windows, Linux, and Mac OS.**

How does Bytecode work in Java?

- When we write a java program, the source code (in the form of .java file) is compiled by *Java compiler* and converted into byte code in the form of a .class file.
- This **compiled byte code is platform-independent code** that can be run on any different computer machine on which JVM *interpreter* is installed. *In simple words, write once, compile and run anywhere (WOCRA).*
- **Java interpreter** translates (converts) bytecode to machine code (native code) line by line during runtime.

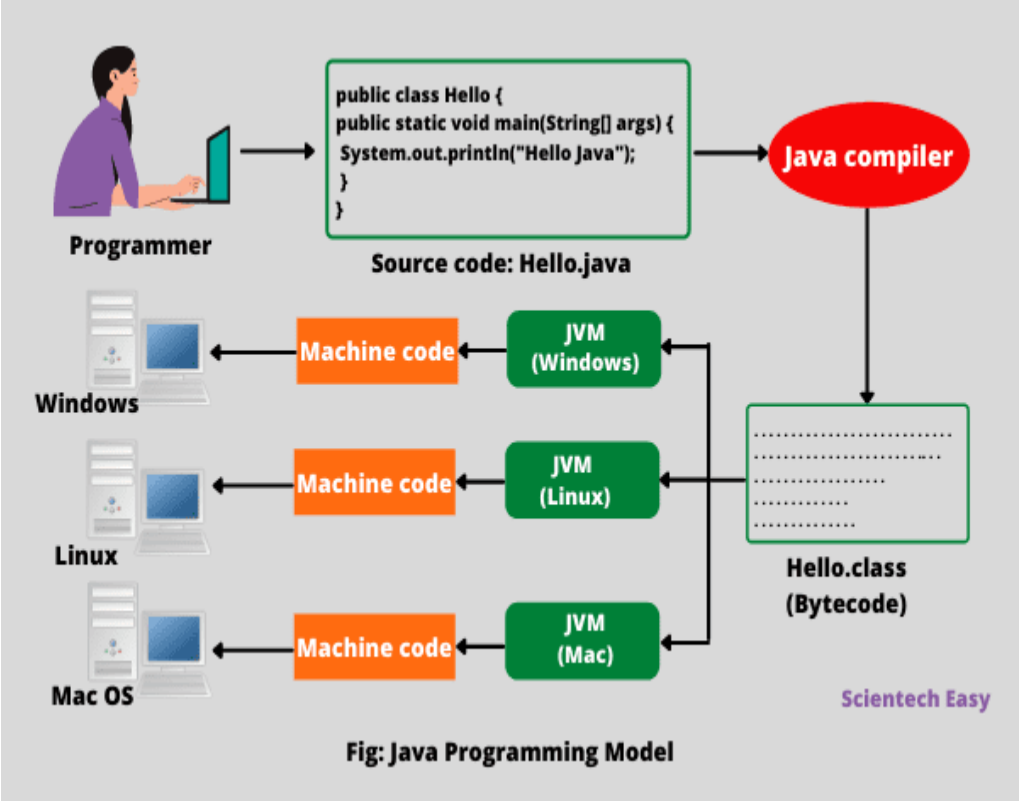


Fig: Java Programming Model

- A **compiler in Java** is a computer program that is used for compiling Java programs. It is platform-independent. **It converts (translates) source code (.java file) into bytecode (.class file).**

Compilation and Execution process

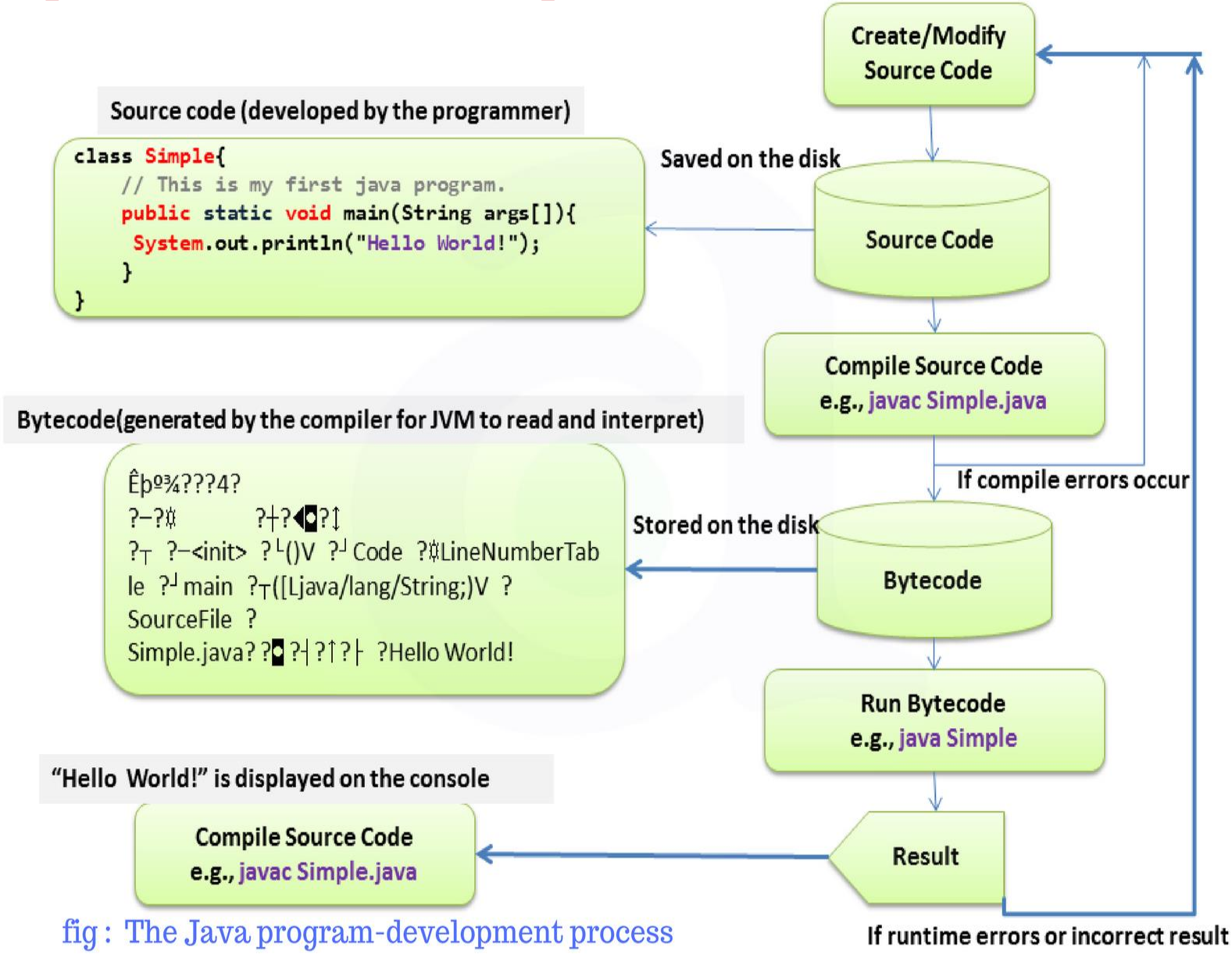


fig : The Java program-development process

Reference Links

POs and Bloom's Levels

https://drive.google.com/file/d/12cDoFByO6W8KRI9Zkg0vEg1UcRL4zDN-/view?usp=sharing

MOOCs:

- http://nptel.ac.in/courses/106106147/
- http://www.nptelvideos.com/java/java_video_lectures_tutorials.php
- http:// www.youtube.com/watch?v=0KL_zftem4g
- http:// www.coursera.org/specializations/object-oriented-programming

OOPJ (22CS307A)

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